

CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 INTRODUCTION:

The **Titan Eye Eyewear E-Commerce Database System** is designed to efficiently manage customers, eyewear products, categories, brands, orders, order items, payments, shipments, and administrators. Functional dependencies help identify the relationships between attributes in each table. Redundancy analysis and data anomaly detection help maintain data consistency and reduce unnecessary duplication. A well-designed database improves data integrity, reduces redundancy, and supports efficient management of the Titan Eye Eyewear e-commerce system.

1. CUSTOMER TABLE:

Attribute	Description
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Contact number
Address	Delivery address
Password	Customer login password

Functional Dependency

Customer_ID → Customer_Name, Email, Phone_Number, Address, Password

Redundancy

Customer information may be duplicated if the same customer registers multiple times or if customer details are stored repeatedly with order information.

Data Anomalies

No major data anomaly is present because customer details are uniquely stored using **Customer_ID**.

2.CATEGORY TABLE:

Attribute	Description
Product_ID	Unique product ID
Product_Name	Eyewear product name
Price	Product price
Stock	Available stock
Category_ID	Category ID
Brand_ID	Brand ID
Frame_Color	Frame color
Lens_Type	Lens type

Functional Dependency

Product_ID → **Product_Name, Price, Stock, Category_ID, Brand_ID, Frame_Color, Lens_Type**

Redundancy

Product details may be duplicated if the same eyewear product is entered multiple times in the database.

Data Anomalies

No major data anomaly is present because each product is uniquely identified by **Product_ID**.

3.BRAND TABLE:

Attribute	Description
Brand_ID	Unique brand ID
Brand_Name	Brand name

Functional Dependency

Brand_ID → **Brand_Name**

Redundancy

Brand information may be repeated for multiple products if the brand name is stored directly in every

product record.

Data Anomalies

No major data anomaly is present because each brand is uniquely identified using Brand_ID.

4.PRODUCT TABLE:

Attribute	Description
Product_ID	Unique product ID
Product_Name	Eyewear product name
Price	Product price
Stock	Available stock
Category_ID	Category ID
Brand_ID	Brand ID
Frame_Color	Frame color
Lens_Type	Lens type

Functional Dependency

Product_ID → **Product_Name, Price, Stock, Category_ID, Brand_ID, Frame_Color, Lens_Type**

Redundancy

Product details may be duplicated if the same eyewear product is entered multiple times in the database.

Data Anomalies

No major data anomaly is present because each product is uniquely identified by **Product_ID**.

5.ORDER TABLE:

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

Functional Dependency

Order_ID → **Customer_ID, Order_Date, Total_Amount, Order_Status**

Redundancy

Order information may be repeated if duplicate order records are created for the same customer.

Data Anomalies

No major data anomaly is present because each order is uniquely identified by Order_ID.

6.ORDER ITEM TABLE:

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

Functional Dependency

Order_Item_ID → Order_ID, Product_ID, Quantity, Price

Redundancy

Product details may be repeated in multiple order item records when the same product is purchased in different orders.

Data Anomalies

No major data anomaly is present because each order item is uniquely identified by Order_Item_ID

7 .PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	UPI/Card/Cash on Delivery
Payment_Status	Paid/Pending

Functional Dependency

Payment_ID → **Order_ID**, **Payment_Method**, **Payment_Status**

Redundancy

Payment information may be duplicated if multiple payment records are created for the same order or transaction.

Data Anomalies

No major data anomaly is present because payment details are uniquely stored using **Payment_ID**.

8. SHIPMENT TABLE

Attribute	Description
Shipment_ID	Unique shipment ID
Order_ID	Order ID
Tracking_Number	Tracking number
Delivery_Status	Delivered/In Transit/Pending

Functional Dependency

Shipment_ID → **Order_ID**, **Tracking_Number**, **Delivery_Status**

Redundancy

Shipment details may be repeated if duplicate shipment records are stored for the same order.

Data Anomalies

No major data anomaly is present because shipment details are uniquely identified using **Shipment_ID**.

10.ADMIN TABLE

Attribute	Description
Admin_ID	Unique admin ID
Admin_Name	Administrator name
Email	Email address
Password	Login password

Functional Dependency

Admin_ID → Admin_Name, Email, Password

Redundancy

Administrator information may be duplicated if identical admin records are stored multiple times.

Data Anomalies

No major data anomaly is present because administrator details are uniquely stored using **Admin_ID**.

CONCLUSION:

The **Titan Eye Eyewear E-Commerce Database System** is designed using appropriate functional dependencies and normalized tables. Each entity stores only relevant information related to customers, products, categories, brands, orders, order items, payments, shipments, and administrators. The use of primary keys and foreign keys reduces data redundancy and prevents insert, update, and delete anomalies. This database design improves data consistency, maintains data integrity, and supports efficient management of the Titan Eye Eyewear e-commerce system. The functional dependencies are based on the primary keys and attributes identified in the Entity Analysis.